

DOCUMENT CONTROL

High-voltage safety and controls

Product-level verification specification for the synthetic Aquila 800 V battery system. Numbered clauses are mandatory unless an explicit logic gate states otherwise.

Table 1. Document control

Field	Value
Document type	Controlled system specification
Authority	Aquila Systems Engineering
Revision / effective date	Rev C / 2026-06-15
Verification evidence	Approved test reports and signed inspection records

Aquila HV safety and monitoring architecture

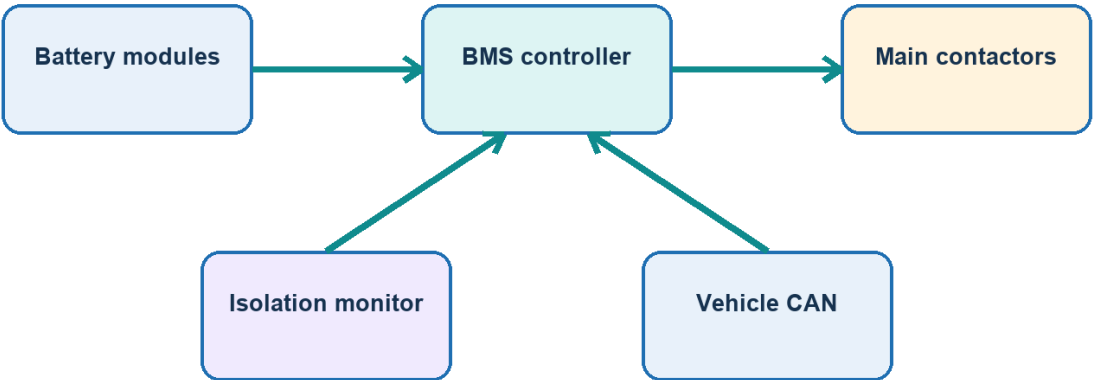


Figure 1. Logical context only; acceptance is defined by the numbered requirements.

Figure 1. System architecture context. The diagram is informative; requirements remain normative.

ELECTRICAL OPERATING ENVELOPE

Electrical operating envelope

MM-REQ-001 - Pack voltage monitoring

The BMS shall continuously monitor pack voltage throughout 400 V to 800 V DC.

- C1:** lower bound coverage shall be at or below 400 V
- C2:** upper bound coverage shall be at or above 800 V

MM-REQ-002 - Cell measurement accuracy

Cell-voltage measurement error shall not exceed +/-5 mV from -20 deg C through +60 deg C.

- C1:** absolute error shall be ≤ 5 mV
- C2:** the verified temperature range shall include -20 deg C
- C3:** the verified temperature range shall include +60 deg C

CRASH ELECTRICAL SAFETY

Crash electrical safety

MM-REQ-003 - Crash disconnect response

On a validated crash input, both main contactors shall open within 20 ms and the DC bus shall fall below 60 V within 5.0 s.

C1: contactor opening latency ≤ 20 ms

C2: DC bus voltage < 60 V by 5.0 s

MM-REQ-004 - Post-impact isolation

At 60 s after impact, measured isolation resistance divided by the measured DC-bus voltage shall be at least 500 ohm/V.

C1: measurement taken at 60 s after impact

C2: calculated isolation ratio ≥ 500 ohm/V

FLUID INTEGRITY AND THERMAL PROTECTION

Fluid integrity and thermal protection

MM-REQ-005 - Coolant circuit integrity

The coolant circuit shall hold 2.0 bar for 30 minutes with no visible leakage.

C1: pressure ≥ 2.0 bar for ≥ 30 min

C2: no visible coolant leakage

MM-REQ-007 - Thermal shutdown and recovery

The BMS shall command shutdown at or above 85 deg C and shall permit recovery only after temperature is below 70 deg C for 60 s.

C1: shutdown asserted at temperature ≥ 85 deg C

C2: recovery requires temperature < 70 deg C continuously for 60 s

VISUAL MARKING AND COMMUNICATIONS

Visual marking and communications

MM-REQ-006 - Service-disconnect hazard marking

Every service-disconnect cover shall carry a readable yellow triangular warning label with a black lightning symbol before the cover can be accessed.

C1: marking design is yellow triangle with black lightning symbol

C2: the marking is present and readable on every service-disconnect cover

MM-REQ-008 - CAN heartbeat endurance

The BMS CAN heartbeat interval shall not exceed 100 ms during a continuous 30-minute run, with zero lost heartbeat messages.

C1: maximum heartbeat interval ≤ 100 ms

C2: zero lost messages during ≥ 30 min

INGRESS PROTECTION AND METROLOGY

Ingress protection and metrology

MM-REQ-009 - Connector J17 ingress protection

Connector J17 shall remain IP67 after 1 m water immersion for 30 minutes, with no moisture inside the rear boot.

C1: immersion depth ≥ 1 m for ≥ 30 min

C2: no moisture inside rear boot

MM-REQ-010 - Instrument traceability

Every measuring instrument used for acceptance data shall be identified by serial number and have calibration valid within 12 months on the test date.

C1: each acceptance measurement identifies an instrument serial

C2: each identified instrument has calibration age ≤ 12 months

CONDITIONAL AND ALTERNATIVE SAFETY LOGIC

Conditional and alternative safety logic

MM-REQ-011 - Drive-away inhibition

If the charge connector is engaged, commanded propulsion torque shall remain zero and vehicle movement shall not exceed 150 mm.

C1: charge-engaged propulsion torque = 0 Nm

C2: charge-engaged vehicle movement $\leq$ 150 mm

MM-REQ-012 - Alternative post-impact electrical protection

After frontal impact, electrical protection is acceptable when either the isolation ratio is at least 500 ohm/V or an isolation fault causes the main contactors to open within 100 ms.

C1: post-impact isolation ratio $\geq$ 500 ohm/V

C2: isolation-fault contactor opening latency $\leq$ 100 ms

Logic rule. MM-REQ-011 is IF_THEN. MM-REQ-012 is ANY_OF: proving either alternative is sufficient, but an untested alternative must still remain visible in the audit trace.